

# QUORUM 1.8 field test protocol — baseline motion gate

Date: 2026-08-07 Governing spec: QUORUM\_BASELINE\_MOTION\_GATE\_SPEC.md §5 (Sam + CODEX reviewed) Decision at stake: 0017 (Proposed → Accepted only if this protocol passes) Locomotive: Otto (9950011). Total track time: roughly 45–60 minutes.

**The order is load-bearing. Stage A runs on the 1.7 currently aboard Otto. Do not flash 1.8 first — the gate makes the bug undemonstrable, and Stage A is the falsifier the whole decision rests on.**

## 0. Safety and operating frame

- All stages run in **MANUAL** — your throttle, your direction. AUTO is never enabled in this protocol.

- Track clear: no other locomotives powered on the Lowline.
- E-stop available on the console throughout.
- Stage A **deliberately corrupts navigation**. Expect DISAGREES,

EVALUATING, possibly NO\_QUORUM. This is the point. Recovery is the normal one: stop, re-declare the start interval, continue.

- Stage C5 (deliberate stall) energizes a held locomotive. Keep PWM  $\leq 30$ , watch telem/current (live on the dashboard since 1.7), and abort the row if current climbs above ~1 A or anything warms.

## 1. Pre-flight checklist (5 min)

[ ] Otto powered, on the main, connected to the broker (dashboard tiles updating).

[ ] Confirm Stage-A firmware is 1.7: state/bootid shows "sketch": "QUORUM\_1\_7" **and** a passing marker line carries "pwm": and "v":.

[ ] Console ready to re-declare a start interval (you will need it after A1).

[ ] Pick your two park spots in advance:

- **MAGNET spot** — sensor directly over a marker magnet. Confirm by parking and reading delta in state/loopstat (dashboard or capture): **|delta|  $\geq 100$**  and steady. Note the sign: **positive = N pole, negative = S pole**. Note the MM number.

- **CLEAR spot** — sensor well away from any magnet: **|delta|  $\leq 5$**  and steady.

[ ] Terminal ready for captures. Every stage uses the same command, only the filename changes:

```
cd ~/esp-loco-control
mosquitto_sub -h 192.168.68.142 \
-t 'ngr/loco/9950011/state/loopstat' -t 'ngr/loco/9950011/mm/marker' \
-t 'ngr/loco/9950011/state/nav' -t 'ngr/loco/9950011/alert' \
-t 'ngr/loco/9950011/telem/voltage' -t 'ngr/loco/9950011/telem/current' \
-v -W 600 > field-records/logs/20260807_<STAGE>.log
```

-w 600 self-terminates after 10 minutes; re-run it fresh per stage. **Start the capture BEFORE the locomotive stops moving, every time.** The corruption self-erases in ~30–60 s of driving; a capture that starts after the dwell proves nothing.

Quick trackside baseline check (run against any capture, live or after):

```
grep -o '"baseline":[0-9]*' field-records/logs/20260807_<STAGE>.log | sort | uniq -c
```

One dominant value = stable. A ladder of values = migration.

## 2. Stage A — pre-fix reproduction, on 1.7 (the falsifier)

### A1 — magnet park (~6 min)

1. Start capture → 20260807\_A1\_prefix-magnet-park.log.
2. Drive Otto normally for  $\geq 60$  s (establishes the healthy baseline in the log).
3. Stop with the sensor **on the MAGNET spot**. Throttle to 0. Confirm  $|\text{delta}| \geq 100$ .
4. Sit  $\geq 120$  s. Watch baseline in loopstat once a minute.
5. Drive off at normal cruise. Keep driving  $\geq 60$  s past ~10 markers.
6. When navigation degrades, let it: do not rescue it early. After the minute, stop, note the nav state, and recover (re-declare interval).
7. Stop the capture (Ctrl-C if before the 10-min timeout).

#### Predictions (1.7):

- During the dwell, baseline in loopstat **\*\*migrates toward the magnet level\*\*** — tens to ~200+ counts within ~60 s — while Otto sits still.
- An mm/marker line appears mid-dwell or at departure with a huge ms and/or extreme drift.
- After departure: wrong-polarity readings, DISAGREE events, miss streak, EVALUATING with a flat score vector; possibly NO\_QUORUM.

**PASS (diagnosis confirmed):** baseline migration  $\geq 40$  counts during the dwell (i.e., beyond the  $\pm 38$  window) with the departure disturbance. **FAIL (diagnosis wrong):** baseline holds within  $\pm 5$  for the whole dwell. → **STOP THE PROTOCOL.** 1.8 is not flashed; bring me the log.

### A2 — control park (~4 min)

Same steps at the **CLEAR spot** (120 s dwell) → 20260807\_A2\_prefix-clear-park.log. **Expected:** baseline steady ( $\pm 3$ ), clean departure, zero DISAGREES. This row proves the effect is the magnet, not the stopping.

## 3. Gate decision

A1 PASS + A2 clean → proceed to Stage B. Anything else → stop, captures to me, no flash.

## 4. Stage B — flash 1.8 and verify (~5 min)

1. Flash from the repo (working tree = committed 1.8), or use the

verified binary (SHA-256 f92a45ef113ef875...):

```
arduino-cli upload -p /dev/cu.usbserial-0001 --fqbn esp32:esp32:esp32 --input-dir /private/tmp/claude-501/-Users-da
```

2. Boot-serial verification, in order — all three lines, not just the

first (the bootid lesson):

```
[BOOT] QUORUM_1_8 - 9950011
[INA] ready
[CAL] 2 s baseline - keep clear of magnets
```

3. Boot **clear of magnets** (the [CAL] instruction is now the only

unconditional baseline authority — that's the design).

4. Set session direction and start interval; drive a few markers; confirm

AGREEs and pwm/v still on the marker line.

The *behavioural* proof of genuine 1.8 is Stage C row 3: a frozen baseline under a big steady delta. Version strings alone don't count.

## 5. Stage C — acceptance matrix on 1.8 (~25 min)

Five parks. Same choreography as A1 for each: capture first → drive ≥ 60 s → park > **70 s** (the u16 saturation threshold is 65.535 s — 70+ makes it unambiguous) → drive off ≥ 60 s → stop capture. One log per row:

| Row | Park condition                                                                         | File suffix | PASS criteria                                                                                                     |
|-----|----------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------|
| C1  | CLEAR spot                                                                             | C1_clear    | baseline ±3 throughout; clean restart; no DISAGREE                                                                |
| C2  | <b>Fringe</b> : edge of a magnet's field, steady $10 \leq  \text{delta}  \leq 35$      | C2_fringe   | baseline ±3 throughout ( <b>this row is why the motion gate beats an excursion gate</b> — 1.7 would have crept)   |
| C3  | <b>N magnet</b> (delta ≥ +100)                                                         | C3_north    | baseline <b>frozen</b> ; saturation checklist below; polarity <b>N</b>                                            |
| C4  | <b>S magnet</b> (delta ≤ −100)                                                         | C4_south    | same, polarity <b>S</b>                                                                                           |
| C5  | Magnet, <b>PWM ~25–30, physically restrained</b> (chock/hand, ≤ 30 PWM, watch current) | C5_stall    | baseline <b>migrates</b> — the documented residual, reproduced on purpose. Not a failure; a boundary made visible |

**Saturation checklist for C3 and C4** (CODEX #3 — check in the log):

- ☐ Exactly **one** mm/marker event for the parked magnet
- ☐ "ms": 65535 (the capped stopwatch)
- ☐ "drift" near 0 (frozen baseline = nothing drifted during the event)
- ☐ Polarity = the pole you parked on (N for C3, S for C4)
- ☐ MM advanced by exactly **one** for that magnet
- ☐ The **next** magnet after departure is a normal AGREE, not rejected

as a phantom

- ☐ No weird station/age behaviour from the old arrival timestamp

Note for C3/C4: the marker event may arrive at *departure* (the event closes when the magnet clears). During the dwell itself the marker line stays silent while an event sits open — that silence is expected.

## 6. Stage D — regression lap (~5 min)

Capture → 20260807\_D\_regression-lap.log. One full lap at normal cruise, plus a short **low-throttle crawl segment** (PWM < 40 over a few markers, to confirm LOW\_PWM-gated markers still navigate).

**PASS:** miss\_streak 0 for the lap; loop\_max\_gap\_ms 33–34-ish and hall\_task\_max\_gap\_ms 2–3 (the 2026-08-06 baseline); crawl markers accepted with timing\_gate:"LOW\_PWM"; pwm/v present throughout.

## 7. Afterwards

Commit all logs to field-records/logs/ (or hand them to me — I'll analyze, commit, and write the field verdict).  
Outcomes:

- **A PASS + C/D PASS** → 0017 promoted to Accepted; field verdict

committed; Station Stop v1 unblocked.

- **A PASS, any C/D row FAIL** → 1.8 stays flashed only if the failure is benign; otherwise reflash 1.7 (archive tag daf468d~ builds it) and we regroup on the spec.

- **A FAIL** → protocol stopped at §3; diagnosis reopened from the A1 log.

## Abort criteria (any stage, any time)

- Current above ~1 A sustained, or anything warm → throttle 0, abort row.
- Runaway or unresponsive console → E-stop.
- Weather/track hazard → stop; captures are cheap, redo the row later.